Home Assistant Native App — UI Component Design Spec

Version: 0.2
Status: Draft
Platform: iOS / macOS (SwiftUI primary; adaptive for iPadOS)
Design System: Maple Design System v1.0

Table of Contents

1. Design Language
2. Design Tokens
3. Component Groups
    - 3.1 Toggle Components
    - 3.2 Slider Components
    - 3.3 Multi-Control Cards
    - 3.4 Read-Only Badges & Tiles
    - 3.5 Picker Components
    - 3.6 Action Button Components
    - 3.7 Media & Stream Components
    - 3.8 Info & Status Components
4. Shared Interaction Patterns
5. Accessibility
6. Error & Unavailable States
7. Entity Detail Sheet

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1. Design Language

Relationship to Maple Design System

This app is part of the Maple ecosystem and inherits Maple's full visual language: typography (Merriweather + Lato), warm neutral surfaces, spacing scale, radius scale, elevation scale, and the single orange accent (`#E8540A`). It is **not** a standalone design system — it extends Maple with an HA-specific semantic color layer for entity state communication.

Any token not defined in this doc defers to Maple's canonical values.

Philosophy

Editorial utility. The app should feel like a well-designed field journal for your home — clear, warm, trustworthy. Not a dark control room, not a consumer smart-home toy. The Merriweather typeface gives the UI editorial weight; the warm neutral base (`#F2F1EF`) keeps it grounded and readable in any light condition.

Aesthetic Direction

- **Light-first.** Warm off-white base (`#F2F1EF`) with white card surfaces. Dark mode is a supported adaptive inversion, not the primary target.
- **Grid-native.** All cards snap to a 4pt baseline grid (Maple `--sp-*` scale). Dashboard layouts use a configurable column grid (2-col default on phone, 3–4 on iPad).
- **State-forward.** Active vs. inactive is the dominant visual signal. Semantic colors carry entity-state meaning; they appear only when entities are active/alerting — otherwise cards are neutral.
- **Maple accent as the app accent.** The orange `#E8540A` is the app's primary interactive color: CTAs, active toggles, progress arcs, FABs. Entity semantic colors (yellow for lights, teal for climate cool, etc.) are additive, not replacements for the app accent.
- **Minimal chrome.** No background gradients. Elevation via Maple's shadow scale only. Icons are SF Symbols (monoline inactive, filled active).

Typography

Merriweather and Lato are bundled as custom fonts in the iOS app target. SwiftUI custom font registration via `Info.plist` + `UIFont.familyNames` .

				Maple

Role	Font	Weight	Size	Analog
Dashboard greeting / section hero	Merriweather	900	28pt	Display LG
Card title / entity name (detail)	Merriweather	700	20pt	Display MD
Area / section header	Lato	700	11pt, UC, +0.1em	Label
Card label (entity name on card)	Lato	700	13pt	Body SM bold
Card value (state readout)	Lato	900	15pt	Body MD bold
Unit / secondary	Lato	300	12pt	Caption
Button	Lato	700	14pt	—
Attribute key/value	Lato	400	13pt	Body SM
Mono (entity IDs, hex values)	Courier New	—	11pt	Mono

Iconography

- SF Symbols exclusively.
- **Inactive:** `.regular` weight, `textMuted` color (`#8A8A8A`).
- **Active:** `.fill` variant, entity semantic accent color.
- Standard card icon size: **24×24pt** (`.medium` symbol scale).
- Detail sheet icon: **56×56pt** on a `surface` rounded square (`radius-lg`) with `shadow-sm` .

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2. Design Tokens

Layer Structure

```
Maple Base Tokens          ← inherited from Maple Design System
  └── HA Semantic Layer    ← entity-state colors defined here
```

```
└── Component Tokens ← composed from both layers above
```

Maple Base Tokens (Native Mapping)

Direct mappings from Maple CSS variables to SwiftUI `Color` definitions:

```
// Surfaces
Color("base")           = #F2F1EF   // app background, inactive fills
Color("surface")        = #FFFFFF   // card backgrounds, inputs
Color("surfaceRaised")  = #FAFAF8   // elevated cards on hover / pressed
Color("overlay")        = #000000CC // modal scrim

// Text
Color("textPrimary")    = #111111
Color("textSecondary")  = #4A4A47
Color("textMuted")      = #8A8A8A
Color("textFaint")      = #C4C4C0
Color("textInverse")    = #FFFFFF

// Accent (Maple orange — primary interactive color)
Color("accent")         = #E8540A
Color("accentHover")    = #C94608
Color("accentDim")      = rgba(232, 84, 10, 0.10)

// Borders
Color("border")         = rgba(0, 0, 0, 0.08)
Color("borderStrong")   = rgba(0, 0, 0, 0.15)

// Semantic
Color("success")        = #2D8A4E
Color("successDim")     = rgba(45, 138, 78, 0.10)
Color("warning")        = #B07D10
Color("warningDim")     = rgba(176, 125, 16, 0.10)
Color("error")          = #C0392B
Color("errorDim")       = rgba(192, 57, 43, 0.10)
Color("info")           = #2874A6
Color("infoDim")        = rgba(40, 116, 166, 0.10)
```

HA Semantic Layer (Entity State Colors)

Entity-state colors layer on top of Maple base. They appear only when an entity is active, alerting, or requires distinct state communication. All are tuned for legibility on `surface` (#FFFFFF) and `base` (#F2F1EF) backgrounds.

```
// Entity State Accents
```

```
Color("entityLight")    = #F59E0B   // lights on (warm amber, readable on white)
Color("entityCool")     = #0D9488   // climate cool, covers open (teal)
Color("entityHeat")     = #EA580C   // climate heat (close to --accent, slightly
Color("entityAlert")    = #DC2626   // locks, alarms, smoke (red)
Color("entityMedia")    = #7C3AED   // media player active (violet)
Color("entitySwitch")   = #2563EB   // switches, generic active (blue)
Color("entityInactive") = #8A8A8A   // same as textMuted, all inactive icons

// Entity State Fills (low-opacity tints on active card backgrounds)
// Each is the entity accent at 8% opacity on --surface
Color("fillLight")      = rgba(245, 158, 11, 0.08)
Color("fillCool")       = rgba(13, 148, 136, 0.08)
Color("fillHeat")       = rgba(234, 88, 12, 0.08)
Color("fillAlert")      = rgba(220, 38, 38, 0.08)
Color("fillMedia")      = rgba(124, 58, 237, 0.08)
Color("fillSwitch")     = rgba(37, 99, 235, 0.08)
```

> **Design rationale:** Entity semantic colors are distinct from Maple's `accent`. The orange `#E8540A` drives all primary UI actions — active toggle track for general switches, CTA buttons, progress arcs. The entity layer only activates for domain-specific semantics (lights = amber, climate = teal, etc.). This keeps the Maple accent coherent as an app-level signal and prevents the entity layer from overwhelming the base identity.

Spacing Scale

Inherits Maple's 4pt base unit scale, expressed as SwiftUI `CGFloat` constants:

```
enum Spacing {
    static let sp1:  CGFloat = 4
    static let sp2:  CGFloat = 8
    static let sp3:  CGFloat = 12
    static let sp4:  CGFloat = 16
    static let sp5:  CGFloat = 20
    static let sp6:  CGFloat = 24
    static let sp7:  CGFloat = 32
    static let sp8:  CGFloat = 40
    static let sp9:  CGFloat = 56
    static let sp10: CGFloat = 80
}
```

Corner Radii

Inherits Maple's radius scale:

```
enum Radius {
```

```
    static let xs:   CGFloat = 6    // inner badges, code snippets
    static let sm:   CGFloat = 10   // slider thumbs, input fields, album art
    static let md:   CGFloat = 16   // sensor tiles, compact cards
    static let lg:   CGFloat = 22   // standard cards
    static let xl:   CGFloat = 32   // sheets, modals
    static let pill: CGFloat = 100  // toggles, pills, buttons
}
```

Elevation / Shadow

Inherits Maple's shadow scale as SwiftUI `.shadow()` modifiers:

```
shadowXS: y: 1, blur: 3,  opacity: 0.06   // inline chips, badges
shadowSM: y: 2, blur: 8,  opacity: 0.07   // cards (default)
shadowMD: y: 4, blur: 16, opacity: 0.09   // hover / pressed lift
shadowLG: y: 8, blur: 32, opacity: 0.12   // sheets, modals
```

Animation Defaults

```
// Toggle/button snap
Animation.easeInOut(duration: 0.12)

// Card state transitions
Animation.easeInOut(duration: 0.22)

// Sheet presentation
Animation.easeInOut(duration: 0.35)

// Spring (scale feedback, slider thumb)
Animation.spring(response: 0.35, dampingFraction: 0.7)
```

---

3. Component Groups

3.1 Toggle Components

Covers: `light` (simple on/off), `switch`, `input_boolean`, `fan` (base toggle)

Anatomy

```
┌──────────────────────────────────────────────┐  surface, radius-lg, shadow-sm
```

```
│  [icon]  Entity Name                [toggle] │  padding: sp4 (H) × sp3 (V)
│          On · 2m ago                         │  textMuted, Caption (Lato 300 12
└──────────────────────────────────────────────┘
```

- **Container:** `surface` (#FFFFFF), `radius-lg`, `shadow-sm`. Active state adds entity `fill-*` background tint.
- **Icon:** 24×24pt SF Symbol. Inactive: `entityInactive` (`#8A8A8A`). Active: entity semantic accent, `.fill` variant.
- **Label:** `textPrimary`, Lato 700 13pt.
- **Secondary text:** `textMuted`, Lato 300 12pt. Shows "On · 2m ago", "Off", or "Unavailable".
- **Toggle:** Native SwiftUI `Toggle`. `.tint` set per entity type.

States

State	Icon	Card Background	Toggle Track
Off	outline, `entityInactive`	`surface` plain	`textFaint` gray
On	`.fill`, entity accent	entity `fill-*` tint	entity accent
Unavailable	outline, `textFaint`	`surface` at 55% opacity	disabled, grayed
Loading	`ProgressView` spinner	`surface` plain	disabled
Error	`exclamationmark.circle.fill`, `error`	`errorDim`	disabled

Interaction

- **Tap:** Optimistic toggle — UI updates immediately. Rolls back with `.notificationOccurred(.error)` haptic if command fails.
- **Long press (400ms) → entity detail sheet.**
- **Hold-to-confirm mode:** For critical entities (security siren, garage door), require 600ms hold before command fires. `accent` progress ring grows around icon during hold.

Variants

Compact tile (2-column grid, card width < 180pt):

```
┌──────────────────────┐
│  [icon]     [toggle] │  top row
│  Entity Name         │  Lato 700 13pt
│  On · 2m ago         │  Lato 300 12pt, textMuted
└──────────────────────┘
```

Entity-specific accent mapping:

Entity	SF Symbol (inactive / active)	Active Accent	Toggle Tint
`light`	`lightbulb` / `lightbulb.fill`	`entityLight`	`entityLight`
`switch`	`powerplug` / `powerplug.fill`	`accent` (#E8540A)	`accent`
`input_boolean`	`switch.2` / `switch.2`	`accent`	`accent`
`fan`	`fan` / `fan.fill`	`entitySwitch`	`entitySwitch`

3.2 Slider Components

Covers: `light` (brightness + color temp), `cover` (position), `fan` (speed), `input_number`, `number`

Anatomy — Standard Slider Card

```
┌──────────────────────────────────────────┐  surface, radius-lg, shadow-sm
│  [icon]  Entity Name                 88%  │  padding: sp4 (H) × sp3 (V)
│  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━  │  6pt track, radius-pill
└──────────────────────────────────────────┘
```

- **Container:** `surface`, `radius-lg`, `shadow-sm`. Active entity fill tint applied when entity is on.
- **Header row:** icon left, entity name (Lato 700 13pt) center-left, value right (Lato 900 15pt, `textPrimary`).

- **Slider track:** 6pt height, `radius-pill`. Filled portion uses entity semantic accent. Unfilled: `border`. Thumb: 24pt circle, `surface` white, `shadow-sm`.
- **Unit:** inline after value ("88%", "2700K", "23°C") in `textMuted`, Lato 300 12pt.

Interaction

- **Drag:** live updates, 150ms trailing debounce before HA service call fires.
- **Tap track:** jump to position, spring animation on thumb.
- **Haptic:** `.impactOccurred(.light)` at 0%, 50%, 100% (or range min/max for `input_number`).
- **Long press header → entity detail sheet.**

Variant — Light: Brightness + Color Temp

```
┌──────────────────────────────────────┐
│ 💡  Desk Lamp                   On   │
│ ☀  ━━━━━━━━━━━━━━━━━━━━━━  88%       │
│ 🌡  ━━━━━━━━━━━━━          2700K     │
└──────────────────────────────────────┘
```

- Two sliders stacked with `sp2` gap.
- Brightness track: left→right gradient `#C4C4C0` (dim) → `#F59E0B` (`entityLight`, bright).
- Color temp track: gradient warm `#FFD6A0` → cool `#C8DCFF`.
- Card height expands to accommodate both rows.

Variant — Light: RGB Color Picker (detail sheet only)

- Circular HSB color wheel, 240pt diameter.
- Brightness via vertical swipe overlay on wheel.
- Current color shown as 44pt filled circle with `border` ring, `shadow-sm`.
- Hex value below, tappable to open `TextField`.

Variant — Cover Position

```
┌──────────────────────────────────────┐
│ 🪟  Living Room Blind           72%  │
```

```
│  ──────────────────────────────────────  │
│  [▲ Open]        [■ Stop]        [▼ Close]    │  ghost pill buttons
└──────────────────────────────────────────┘
```

- Slider 0 (closed) → 100% (open). Track fill: `entityCool`.
- Three pill buttons (Lato 700 12pt, `borderStrong` ghost style). Stop enabled only while cover is moving.

Variant — Fan Speed

- Slider maps 0–100% or preset mode stops (discrete tick marks on track).
- `fan.fill` icon rotates continuously via `rotationEffect` + `.repeatForever` animation while fan is on. Paused when off.

3.3 Multi-Control Cards

Covers: `climate`, `media_player`, `cover` (compound — see §3.2)

3.3.1 Climate Card

```
┌──────────────────────────────────────────┐
│  🌡️  Living Room AC          [HEAT pill]      │  fillHeat card tint
│                                          │
│        22.4°          →          24°         │
│        Merriweather 700 32pt               │
│        textMuted, Caption (current/target)  │
│                                          │
│  [❄ Cool]  [🔥 Heat]  [♻ Auto]  [⏸️ Off]    │  mode strip
└──────────────────────────────────────────┘
```

- **Container:** `surface`, `radius-lg`, `shadow-sm`. Background tints to `fillHeat` or `fillCool` based on active mode.
- **Header:** entity name (Lato 700 13pt) left. Active mode pill right: mode accent foreground on `*Dim` background (e.g., "HEAT" in `entityHeat` on `fillHeat`).
- **Temperature display:** current temp large (Merriweather 700 32pt) → `→` arrow → target temp. Tap target temp to reveal inline +/– stepper.
- **Mode strip:** horizontal buttons, `radius-pill`. Active: filled entity accent, `textInverse`. Inactive: `base` fill, `textMuted`.

Mode → accent mapping:

Mode	Accent	Background Fill
Cool	`entityCool`	`fillCool`
Heat	`entityHeat`	`fillHeat`
Auto	`accent` (#E8540A)	`accentDim`
Off	`textMuted`	none

3.3.2 Media Player Card

```
┌──────────────────────────────────────────────┐
│  [48×48 art]  Song Title              ···    │
│  radius-sm    Artist · Source                │
│                                              │
│  [⏮]    [⏸/▶]    [⏭]      🔊 ──────────     │
└──────────────────────────────────────────────┘
```

- **Container:** `surface`, `radius-lg`, `shadow-sm`. Playing state adds `fillMedia` tint.
- **Album art:** 48×48pt, `radius-sm`. Falls back to `music.note` in `entityMedia`.
- **Title:** Lato 700 15pt, `textPrimary`. Artist + source: Lato 300 12pt, `textMuted`.
- **Transport:** play/pause 28pt dominant. Prev/next 20pt. `.impactOccurred(.medium)` haptic on all.
- **Volume:** 96pt compact slider, right-aligned. `entityMedia` accent fill. Speaker icon tap-to-mute.
- **Long press → detail sheet** with seek bar (Maple arc primitive), source picker, speaker groups.

3.4 Read-Only Badges & Tiles

Covers: `sensor`, `binary_sensor`, `person`, `device_tracker`

Sensor Tile

```
┌──────────────────────┐   surface, radius-lg, shadow-sm, square aspect
```

```
│  [icon]              │  top-left, 20pt, textMuted
│                      │
│  21.4                │  Merriweather 700 28pt, textPrimary
│  °C                  │  Lato 300 12pt, textMuted
│                      │
│  Bedroom Temp        │  Lato 700 11pt UC, textMuted, bottom
└──────────────────────┘
```

The large Merriweather numeral is the tile's signature — editorial weight on a read-only value. No controls, no chrome.

Binary Sensor Tile

```
┌──────────────────────┐
│  [icon]  OPEN        │  icon 20pt + state pill
│                      │
│  Front Door          │  Lato 700 13pt
│  2m ago              │  Lato 300 12pt, textMuted
└──────────────────────┘
```

- State pill: Lato 700 10pt UC, `radius-pill`. Active: entity accent background, `textInverse`. Clear: `base` background, `textMuted`.
- Icon and pill tint by device class:

Device Class	Icon (off / on)	Active Accent
`motion`	`figure.stand` / `figure.walk`	`entityLight`
`door` / `window`	`door.closed` / `door.left.hand.open`	`entityLight`
`garage_door`	`garage.closed` / `garage.open`	`entityLight`
`smoke`	`smoke` / `smoke.fill`	`entityAlert`
`moisture`	`drop` / `drop.fill`	`entitySwitch`
`battery`	`battery.25` / `battery.100`	`entityAlert` (low) / `success`
`connectivity`	`wifi.slash` / `wifi`	`success`
`plug`	`powerplug` / `powerplug.fill`	`entitySwitch`

Person / Device Tracker Tile

```
┌──────────────────────┐
│  [avatar]  🏠        │  40pt circle, home badge bottom-right
│                      │
│  Zubair              │  Merriweather 700 15pt
│  Home · 4m ago       │  Lato 300 12pt, textMuted
└──────────────────────┘
```

- Avatar circle border: `entityCool` when home, `textFaint` when away.
- Home badge: `house.fill` (`entityCool`) or `location` (`textMuted`), 14pt on white 20pt circle, `shadow-xs`.

3.5 Picker Components

Covers: `input_select`, `select`, `input_datetime`

Select / Input Select

```
┌──────────────────────────────────────────┐
│  [icon]  Entity Name            Option ▾  │
└──────────────────────────────────────────┘
```

- Same container as toggle card. Neutral background — no active fill (value is always a string, not binary).
- Value: right-aligned, Lato 400 13pt, `textMuted`. Chevron: `chevron.down` 12pt, `textFaint`.
- **Tap:** native SwiftUI `Menu` (iOS) or `Picker .menuStyle` (macOS). Active option: `checkmark` in `accent`.
- **Long press → detail sheet** with searchable list when options > 8.

Input Datetime

```
┌──────────────────────────────────────────┐
│  📅  Watering Schedule          07:30 AM  │
│                                  Mon-Fri  │
└──────────────────────────────────────────┘
```

- Time and recurrence right-aligned, stacked. Lato 700 13pt + Lato 300 12pt.
- **Tap:** `.sheet` with native `DatePicker` in `.wheel` style, `accentColor` set to `accent`.

3.6 Action Button Components

Covers: `script`, `scene`, `button`, `input_button`, `timer`

Action Button Card

```
┌──────────────────────┐  surface, radius-lg, shadow-sm
│                      │
│       [icon]         │  32pt, center
│                      │
│    Entity Name       │  Lato 700 13pt, textPrimary, center
│    Last run 4m ago   │  Lato 300 12pt, textMuted, center
│                      │
└──────────────────────┘
```

- Square or landscape flex aspect.
- **Tap:** scale to 0.94 spring, `.impactOccurred(.medium)` haptic.
- **Running state:** icon → `ProgressView` spinner, `accentDim` background tint.
- **Scene card:** icon `sparkles`. `accent` brief tap flash, then returns neutral (no persistent fill state — scenes are fire-and-forget).

Timer Card

```
┌──────────────────────────────────────────┐
│  ⏱  Irrigation Timer              03:42  │  Lato 900 15pt
│  ━━━━━━━━━━━━━━━━━━━━━━━━                │  4pt progress bar, accent fill
│  [▶ Start]    [⏸ Pause]    [× Cancel]    │   ghost pill buttons
└──────────────────────────────────────────┘
```

- Countdown pulses gently (opacity 1.0 → 0.5, 800ms loop) while active.
- Progress bar: 4pt height, `radius-pill`, `accent` fill.
- Control buttons: Lato 700 12pt, `borderStrong` ghost pills. Disabled states applied contextually.

3.7 Media & Stream Components

Covers: `camera` , `image`

Camera Card

```
┌──────────────────────────────────────────────┐  bg black, radius-lg
│                                              │
│   [16:9 stream or still — fills card]        │
│                                              │
│▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒│  .ultraThinMaterial blur bar
│  📷 Front Door              🔴 LIVE  ⤢       │
└──────────────────────────────────────────────┘
```

- **Container:** black, `radius-lg` . No inner padding on image area.
- **Overlay bar:** bottom, `.ultraThinMaterial` . Entity name left (Lato 700 12pt, `textInverse`). LIVE badge + fullscreen right.
- **LIVE badge:** 8pt `entityAlert` dot + “LIVE” Lato 700 10pt UC, `textInverse` . Hidden for still cameras.
- **Tap:** stream → fullscreen player. Still → lightbox + refresh + timestamp.
- **Still auto-refresh:** configurable interval, default 30s. Stale image fades to 60% opacity while reloading.

Image Entity Card

- Simplified camera card, no LIVE badge or stream controls.
- Tap → lightbox. Last-updated timestamp overlay bottom-right (Lato 300 11pt, `textInverse`).

3.8 Info & Status Components

Covers: `automation` , `update`

Automation Card

```
┌──────────────────────────────────────────────┐
│  ⚙  Motion Lights Auto          [toggle]     │
│     Last run: 12m ago                        │  Lato 300 12pt, textMuted
```

```
│     [▶ Trigger now]                          │  ghost pill, Lato 700 12pt
└──────────────────────────────────────────────┘
```

- Extends toggle card (§3.1). Toggle `.tint` uses `accent` (not entity semantic — automation is an app-level concept).
- **Last triggered:** Lato 300 12pt, `textMuted`, below entity name.
- **Trigger button:** `borderStrong` ghost pill. Hidden when automation is off.

Update Card

```
┌──────────────────────────────────────────────┐
│  ↑  Home Assistant            2025.3.1       │  installed, textMuted
│     → 2025.4.0 available                     │  available, textPrimary
│                                              │
│  [Install Update]                            │  accent CTA, radius-pill, full-w
└──────────────────────────────────────────────┘
```

- **Install button:** Maple primary button style (`accent` fill, `radius-pill`, Lato 700 14pt, `textInverse`). `ProgressView` inline while installing.
- **Up-to-date state:** `checkmark.circle.fill` in `success`, "Up to date", Lato 400 13pt, `textSecondary`.

---

4. Shared Interaction Patterns

Optimistic Updates

All control entities update the UI immediately on user action. If the WebSocket command times out (>5s) or returns an error, the UI rolls back to the prior state and shows a transient error toast.

Long Press → Entity Detail

Every card supports long-press to open the Entity Detail Sheet (§7). Minimum press duration: 400ms. Visual feedback: card scales to 0.97 with spring animation on press start.

Pull to Refresh

Dashboard scroll view supports pull-to-refresh, triggering a full state reload via `get_states`. Refresh indicator tinted `accent`.

Contextual Menus

Right-click (macOS) or long-press (iOS) on any card shows:

- **Open Detail** — same as long press
- **Go to Area** — navigates to the area containing this entity
- **Customize Card** — label override, icon override, size config
- **Copy Entity ID** — dev mode only

Drag to Reorder

During long press: 0–400ms nothing, 400ms drag handle appears (card lifts to `shadow-md`), 800ms context menu fires. Releasing 400–800ms enters reorder mode.

---

5. Accessibility

Requirement	Implementation
Dynamic Type	All text uses SwiftUI `Font` with custom font registration. Cards expand via intrinsic size.
VoiceOver	`accessibilityLabel`: "{entity name}, {state}". `accessibilityHint` on controls: "Double-tap to toggle", "Swipe to adjust".
Reduce Motion	Spring scale animations → `.easeInOut(duration: 0.15)` cross-fade. Fan rotation disabled. Arc progress uses opacity only.
Contrast	All Maple text colors meet WCAG AA (4.5:1) on `surface`. Entity accents meet 3:1 minimum for non-text icon use on `surface`.
Minimum Touch Target	All interactive elements ≥ 44×44pt. `.contentShape(Rectangle())` applied to cards.
Switch Control	Traversal order per card: icon → label → primary control → secondary button.
Color Alone	State is never communicated by color alone. State text ("On", "Off", "Open", "Unavailable") always accompanies color signals.

---

6. Error & Unavailable States

Unavailable

- Icon: outline variant, `textFaint`.
- Entity name: `textMuted`. Value: "Unavailable", `textFaint`, Lato 300 12pt.
- Card: `surface` at 55% opacity. No shadow (flat, recessed feel).
- No interactive controls. Tap is no-op. No haptic.

Unknown

- Same as Unavailable. State text: "Unknown". `questionmark.circle` badge 12pt overlaid bottom-right of main icon.

Connection Error

- Toast slides down from top: `surface`, `shadow-lg`, `radius-lg`.
- `exclamationmark.circle.fill` in `error` left, "Connection lost — retrying…" Lato 700 13pt, `ProgressView` right.
- Cards remain at last-known state, dimmed by `base` at 25% opacity overlay.

Command Failed

- Transient toast from bottom: `error` background pill, "Command failed", Lato 700 13pt, `textInverse`. Auto-dismiss 2.5s.
- Card rolls back with `.easeInOut(duration: 0.22)`.
- Haptic: `.notificationOccurred(.error)`.

---

7. Entity Detail Sheet

Full-height modal sheet (`.presentationDetents([.large])`). Drag-to-dismiss enabled. Sheet background: `base` (#F2F1EF) — visually distinct from the `surface` card layer beneath.

Layout

```
┌──────────────────────────────────────────────┐
```

```
│  ────────────────────                [×] │  drag indicator + close (Lato 700 1
│                                          │
│  ┌──────┐                                │
│  │ icon │  Entity Name                   │  Merriweather 700 22pt
│  │56×56 │  entity.id · Area Name         │  textMuted, Courier 11pt / Caption
│  └──────┘                                │
│                                          │
│  ────────────────────────────────────────│  border separator
│                                          │
│  [primary controls — entity-specific]    │
│                                          │
│  ────────────────────────────────────────│
│  ATTRIBUTES                              │  Lato 700 11pt UC, textMuted
│  key          value                      │  Lato rows, 13pt
│                                          │
│  ────────────────────────────────────────│
│  HISTORY              [Last 24h ▾]       │
│  [sparkline / chart]                     │
│                                          │
│  ────────────────────────────────────────│
│  RELATED AUTOMATIONS                     │
│  [automation rows]                       │
└──────────────────────────────────────────┘
```

Detail Icon

56×56pt filled SF Symbol, entity semantic accent color, on `surface` rounded square (`radius-lg` , `shadow-sm`).

Primary Controls by Entity Type

Entity	Controls
`light`	Toggle + brightness slider + color temp slider + RGB wheel (if supported)
`climate`	Temp +/– stepper (±0.5° step) + mode picker + fan mode picker + humidity readout
`media_player`	Seek bar (Maple arc for elapsed %) + transport + source picker + volume + speaker groups
`cover`	Position slider + Open / Stop / Close buttons
`sensor`	Full history chart, selectable range

`camera`	Full-width stream or still, snapshot button, motion toggle (if available)
`automation`	Toggle + last triggered + trigger button + condition/action summary
`timer`	Full controls + progress arc (Maple arc primitive)

History Chart

- **Numeric sensors:** line chart. `entitySwitch` stroke, `accentDim` area fill. Time axis: Lato 300 11pt, `textFaint`.
- **Binary sensors:** step chart. `entityLight` fill for on-periods.
- **Range picker:** segmented control — 1h / 24h / 7d / 30d.
- **Tap crosshair:** exact value + timestamp in tooltip pill (`surface`, `shadow-sm`, `radius-pill`).

Attributes Table

- Key: Lato 700 13pt, `textSecondary`. Value: Lato 400 13pt, `textMuted`, right-aligned.
- Boolean values: `checkmark` / `minus` SF Symbol (not text strings).
- Tap row → copy value to clipboard + `.impactOccurred(.light)`.

Related Automations

- List of automations referencing this `entity_id`. Compact automation card layout per row (§3.8).
- Empty state: “No automations reference this entity.” Lato 300 13pt, `textMuted`, centered.